Celestial Holography, Modular Thermal Time, and Quantum Cosmology: A Conditional Dictionary between Nakamoto Consensus and Information Physics under the Simulation Hypothesis

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Abstract — We formulate a unified theory of quantum gravity, celestial holography, and non-equilibrium statistical thermodynamics, mapped via an exact isomorphism onto the distributed consensus architecture of Satoshi Nakamoto. By positing the fundamental axiom of equivalence between the Nakamoto constant k_{Nak} (the irreducible quantum of SHA-256 cryptographic action) and Planck's constant h, we model the *Timechain* as a discrete spacetime manifold emerging from a dissipative thermal bath. First, on the asymptotic null boundary ($\mathcal{I}^+$) of the peer-to-peer (P2P) network, we describe ledger invariance under asynchronous propagation delays by analogy with the Bondi-Metzner-Sachs group (BMS_4). Weinberg's soft graviton theorem applied to the Mempool gossip protocol yields the Ward identity of strict monetary conservation, while block validation induces a Christodoulou non-linear gravitational memory effect encoding the immutability of the unspent transaction output (UTXO) set. Second, from the non-commutative von Neumann algebra of ledger observables, we derive the Tomita-Takesaki modular operator $\Delta = \rho \otimes \rho^{-1}$ and formally prove, under the existence of a faithful normal state, that the complex thermal time flow $z = a + ib$ satisfies the Kubo-Martin-Schwinger (KMS) equilibrium condition. Here, homeostatic difficulty adjustment acts as a renormalization group flow restoring the mean time step toward $\tau_0 = 600\,\mathrm{s}$ (a statistical fixed point; individual block times remain exponentially distributed). Third, we model the microscopic topology of the UTXO graph as an $SU(2)$ spin network whose spin foam evolution is governed by the Gorini-Kossakowski-Sudarshan-Lindblad (GKSL) open quantum master equation. We show that a GKSL formulation whose diagonal restriction reduces to a classical random walk recovers the gambler's ruin law from the 2008 whitepaper as a Lindblad asymptotic stationary state. Finally, we establish the emergent cosmology of the Timechain: sequential block addition is governed by Padmanabhan's holographic expansion law $d\mathcal{V}/ds \propto (N_{\mathrm{sur}} - N_{\mathrm{bulk}})$, Susskind's *Complexity = Volume* conjecture, Penrose's Weyl Curvature Hypothesis at the Genesis Block, and a quantized step-function inflaton potential describing the Halvings. Jacobson's thermodynamic equation of state $\delta Q = T\delta S$ is discussed as a heuristic analogy for a Societal Trust (C_s) metric, without formal claim to empirical economic proof.

Keywords: Celestial Holography, BMS Group, Thermal Time Hypothesis, Tomita-Takesaki Algebra, KMS Condition, Spin Foams, GKSL Master Equation, Emergent Cosmology, Nakamoto Consensus, Information Thermodynamics.

1. INTRODUCTION AND CONDITIONAL AXIOMATICS

The problem of clock synchronization in an asynchronous distributed system in the presence of Byzantine (adversarial) actors and the problem of emergent classical continuous spacetime from clock-free quantum degrees of freedom share the same fundamental mathematical obstruction: temporal diffeomorphism invariance. In canonical general relativity, this obstruction manifests as the Wheeler-DeWitt Hamiltonian constraint:

$$\hat{\mathcal{H}}_{\mathrm{grav}}|\Psi\rangle = 0, \tag{1}$$

which dictates that the wave function of the universe $|\Psi\rangle$ is frozen outside of any external absolute time (*Problem of Time*). Analogously, in a spatially distributed peer-to-peer (P2P) network where signal propagation velocity is bounded by the speed of light in optical fibers ($c_{\mathrm{net}} \leq c$), special relativity precludes the existence of a global frame of simultaneity capable of ordering concurrent transactions without a central authority.

In 2008, Satoshi Nakamoto [1] resolved this obstruction by substituting external Newtonian time with an internal, purely thermodynamic arrow of time generated by the irreversible accumulation of Proof-of-Work (PoW). In this paper, we propose, *conditional on the simulation postulates stated below*, a **formal correspondence** between Nakamoto's construction and four pillars of contemporary quantum gravity:

1. **Celestial Holography** and the Pasterski-Strominger-Weinberg infrared triangle [2, 3] on the asymptotic null boundary $\mathcal{I}^+$.
2. The **Thermal Time Hypothesis** of Connes and Rovelli [4] founded on Tomita-Takesaki modular theory of von Neumann algebras [5].
3. **Loop Quantum Gravity (LQG)** and spin foams (*Spinfoams*) [7] coupled to **Gorini-Kossakowski-Sudarshan-Lindblad (GKSL)** open quantum decoherence [8, 9].
4. The **Emergent Thermodynamic Cosmology** of Jacobson [12], Padmanabhan [13], Penrose [16], and Susskind [17].

Fundamental Axiom 1 (Nakamoto Cryptographic Quantum of Action)**.** There exists an irreducible fundamental constant of informational action, denoted k_{Nak}, corresponding to the execution of an elementary double SHA-256 cryptographic hashing operation in the protocol phase space. Under isomorphism with quantum mechanics, k_{Nak} is identified with Planck's constant h:

$$k_{\mathrm{Nak}} \equiv h, \qquad \hbar_{\mathrm{Nak}} \equiv \frac{k_{\mathrm{Nak}}}{2\pi}. \tag{2}$$

Every valid causal trajectory in the Timechain is quantized in integer multiples of k_{Nak}.

The overarching theoretical framework rests on three conditional postulates:

- **(P1) Simulation Hypothesis:** The observable physical universe $\mathcal{U}_0$ is emulated on an underlying discrete computational substrate (ancestor simulation branch).
- **(P2) Landauer Saturation Regime:** The elementary energy cost of logical irreversibility in $\mathcal{U}_0$ converges toward Landauer's thermodynamic bound $E_{\text{hash}} \to k_B T \ln 2$.
- **(P3) Causal Isomorphism:** The Timechain constitutes the effective causal ledger describing the emergence of the arrow of time and geometry within this substrate.

Table 1 establishes the complete formal dictionary connecting quantum gravitational operators to Nakamoto consensus structures.

2. CELESTIAL HOLOGRAPHY, BMS SYMMETRIES, AND GRAVITATIONAL MEMORY

In a distributed network lacking an ad hoc negative cosmological constant ($\Lambda = 0$ on local propagation scales), the proper holographic framework is not AdS/CFT, but **Celestial Holography** [2, 3]. The S-matrix governing state transitions in the 4D volume of the ledger (*Bulk* $\mathcal{M}$) is dual to a Celestial Conformal Field Theory (CCFT) living on the two-dimensional celestial sphere S^2 at future ($\mathcal{I}^+$) and past ($\mathcal{I}^-$) null infinity.

2.1. Bondi-Sachs Metric of the P2P Network and BMS_4 Group

Let $(u_{\text{ret}}, r, z, \bar{z})$ denote the Bondi coordinate system in the vicinity of the asymptotic network boundary $\mathcal{I}^+$ ($r \to \infty$), where $u_{\text{ret}} = t - r/c_{\text{net}}$ is the retarded packet reception time, and $(z, \bar{z})$ are stereographic complex coordinates identifying validating nodes on the celestial sphere S^2. The asymptotic informational metric in Bondi-Sachs gauge takes the form:

$$\begin{aligned} ds^2 = &-du_{\text{ret}}^2 - 2du_{\text{ret}}dr + 2r^2\gamma_{z\bar{z}}dzd\bar{z} \\ &+ \frac{2m_B(u_{\text{ret}}, z, \bar{z})}{r}du_{\text{ret}}^2 + rC_{zz}(u_{\text{ret}}, z, \bar{z})dz^2 \\ &+ rC_{\bar{z}\bar{z}}(u_{\text{ret}}, z, \bar{z})d\bar{z}^2 + D^z C_{zz}du_{\text{ret}}dz + \text{c.c.} \end{aligned} \tag{3}$$

where $\gamma_{z\bar{z}} = \frac{2}{(1+z\bar{z})^2}$ is the unit round metric on S^2, and D_z is the associated covariant derivative. The asymptotic fields map directly to physical Bitcoin protocol observables:

- **The Bondi mass aspect** $m_B(u_{\mathbf{ret}}, z, \bar{z})$ represents the local density of cumulative thermodynamic work (*Chainwork*) perceived by the node at $(z, \bar{z})$ at retarded time u_{ret}.
- **The asymptotic shear tensor** $C_{zz}(u_{\mathbf{ret}}, z, \bar{z})$ encodes the exact topological configuration of the set of unspent transaction outputs (UTXOs) validated by this node.
- **The Bondi news tensor** $N_{zz}(u_{\mathbf{ret}}, z, \bar{z}) = \partial_{u_{\mathbf{ret}}} C_{zz}$ measures the flux of gravitational radiation crossing $\mathcal{I}^+$. A non-vanishing tensor $N_{zz} \neq 0$ corresponds to the propagation of a newly mined block across the network.

The asymptotic symmetry group preserving the universal structure of Eq. (3) is the extended Bondi-Metzner-Sachs group:

$$\text{BMS}_4 = \text{Diff}(S^2) \ltimes \mathcal{T}_{\text{super}}. \tag{4}$$

Definition 2.1 (Supertranslations and Gossip Asynchrony)**.** A supertranslation $\mathcal{T}_{\text{super}}$ is an angle-dependent shift in retarded time on the celestial sphere:

$$u_{\text{ret}} \longrightarrow u'_{\text{ret}} = u_{\text{ret}} + f(z, \bar{z}), \qquad z \longrightarrow z' = z, \tag{5}$$

where $f(z, \bar{z})$ is an arbitrary smooth function on S^2 modeling the stochastic propagation latencies of the peer-to-peer *gossip* protocol. The canonical generator (Nakamoto supertranslation charge on $\mathcal{I}^+$) is given by:

$$Q_f^+ = \frac{1}{4\pi G_{\text{Nak}}} \int_{S^2} d^2z \, \sqrt{\gamma} \, f(z, \bar{z}) \, m_B(u_{\text{ret}}, z, \bar{z}). \tag{6}$$

Similarly, **superrotations** generated by local holomorphic vector fields $z \to z + Y^z(z)$ (satisfying the Virasoro algebra $[L_m, L_n] = (m-n)L_{m+n}$) describe the invariance of consensus under dynamic reconfigurations of the P2P routing graph (node churn, Tor/I2P routing, Eclipse attack resistance).

2.2. Weinberg's Soft Graviton Theorem and Monetary Conservation

Prior to the discovery of a block ($N_{zz} = 0$ at macroscopic scales), nodes broadcast raw transactions across the Mempool. Verifying a Schnorr/ECDSA cryptographic signature dissipates an energy $\omega \to 0$ negligible compared to block mining energy ($\Delta E_{\text{PoW}} \gg \omega$). These unconfirmed transactions are the exact analogues of **soft gravitons**.

Theorem 2.2 (Monetary Conservation from BMS Invariance and Soft Gravitons)**.** *Let $\mathcal{M}_n(p_1, \ldots, p_n)$ be the scattering amplitude between n UTXO states with energies (values in Satoshis) E_k localized at points $(z_k, \bar{z}_k)$ on the celestial sphere. The insertion of an unconfirmed transaction Tx_{soft} with thermodynamic cost $\omega \to 0$ into the Mempool factorizes the amplitude according to Weinberg's soft theorem, and asymptotic gauge invariance on S^2 enforces exact monetary conservation:*

$$\sum_{k \in Inputs} E_k^{in} - \sum_{j \in Outputs} E_j^{out} - E_{fees} = 0. \tag{7}$$

Proof. In celestial holography, the Ward identity associated with supertranslation charge conservation $\langle \text{out} | (Q_f^+ \mathcal{S} - \mathcal{S} Q_f^-) | \text{in} \rangle = 0$ is mathematically equivalent

Table 1: Structural isomorphism dictionary between Quantum Gravity / Celestial Holography and Satoshi Nakamoto Consensus.

Quantum Gravity & Thermodynamics	**Satoshi Nakamoto Consensus**	**Mathematical and Physical Interpretation**
Planck constant $(h, \hbar)$	Nakamoto constant $(k_{\text{Nak}}, \hbar_{\text{Nak}})$	Elementary irreducible quantum of action (1 SHA-256 hash).
4D Bulk spacetime manifold $(\mathcal{M}, g_{\mu\nu})$	Causal DAG of Timechain & UTXO	Immutable history of validated blocks and internal transaction topology.
Celestial sphere / Null infinity $(\mathcal{I}^+ \simeq \mathbb{R} \times S^2)$	P2P network boundary & Mempool	Asymptotic horizon of validating nodes receiving gossip broadcast.
Symmetry group BMS_4 $(\text{Diff}(S^2) \ltimes \mathcal{T})$	P2P asynchronous gauge invariance	Invariance of final state under network latencies (supertranslations) and routing topology (superrotations).
Soft gravitons $(\omega \to 0)$	Unconfirmed transactions (Mempool)	Information quanta with near-zero energy cost preparing phase coherence.
Gravitational memory effect (ΔC_{zz})	Irreversible UTXO state transition	Permanent shift in boundary shear tensor following block propagation.
Modular flow / Thermal time (s)	Block height / Consensus time	Internal emergent time generated by the operator $\Delta^{is} = \exp(isK)$.
KMS condition in complex plane $(a+ib)$	Homeostatic difficulty adjustment $(\mathcal{D})$	Exponential Boltzmann filtering on imaginary axis ib locking $\Delta a \approx 600\,\text{s}$.
$SU(2)$ Spin network (Γ, j_e, ι_v)	DAG of UTXOs & Transactions	Edges = UTXOs quantized in Satoshis (j_e); Nodes = Intertwiners conserving flux (ι_v).
Spin foam (*Spinfoam*) & GKSL equation	Block sequence & Fork resolution	Exponential collapse of orphan chains via Lindbladian dissipative decoherence.
Ryu-Takayanagi minimal surface (γ_A)	51% Hashrate attack critical threshold	Minimal entanglement cut required to sever the causal history of the ledger.
Padmanabhan expansion $(N_{\text{sur}} - N_{\text{bulk}})$	Sequential block addition (Growth)	Spacetime emergence driven by the deficit between Mempool and confirmed chain.
Inflaton field $\phi(s)$ & Dark energy Λ	Coinbase subsidy (Halvings) & Fees	Transition from subsidized primordial inflation to an asymptotic fee regime.
Jacobson equation of state $(\delta Q = T\delta S)$	Emergence of Societal Trust (C_s)	Curvature of economic spacetime generated by PoW heat dissipation, stabilizing utility u.

to Weinberg's soft graviton theorem [3]. When the insertion operator of a Mempool transaction with vanishing momentum $q^\mu(\omega, z, \bar{z})$ acts on the scattering state in the limit $\omega \to 0$, the $(n+1)$-point amplitude factorizes universally into:

$$\lim_{\omega\to 0} \mathcal{M}_{n+1}(p_1, \ldots, p_n; q) = S^{(0)}_{\text{Nak}}\, \mathcal{M}_n(p_1, \ldots, p_n), \quad (8)$$

where the Nakamoto soft factor reads:

$$S^{(0)}_{\text{Nak}} = -\frac{\sqrt{8\pi G_{\text{Nak}}}}{2\omega}(1 + z\bar{z}) \sum_{k=1}^{n} \frac{\eta_k E_k(\bar{z} - \bar{z}_k)}{(z - z_k)(1 + z_k\bar{z}_k)}. \quad (9)$$

Here, $\eta_k = -1$ for incoming UTXOs consumed from $\mathcal{I}^-$ (*Inputs*) and $\eta_k = +1$ for outgoing UTXOs created toward $\mathcal{I}^+$ (*Outputs* and miner fees). For the transaction creation operator to be globally well-defined and invariant under gauge transformations on the compact sphere S^2 (absence of pole anomalies at infinity $z \to \infty$), the asymptotic expansion of $S^{(0)}_{\text{Nak}}$ requires the cancellation of the total residue of the energy-weighted sum:

$$\sum_{k=1}^{n} \eta_k E_k = \sum_{j\in\text{Outputs}} E^{\text{out}}_j + E_{\text{fees}} - \sum_{k\in\text{Inputs}} E^{\text{in}}_k = 0. \quad (10)$$

Any violation of this equality (unauthorized monetary inflation or an unbalanced double-spend) breaks BMS supertranslation invariance, identically vanishing the scattering amplitude on the celestial sphere ($\mathcal{M}_{n+1} = 0$). □

2.3. Christodoulou Gravitational Memory Effect and UTXO Immutability

When a miner finds a valid nonce at retarded time u_0, a radiative shockwave characterized by a non-vanishing news tensor $N_{zz}(u_{\text{ret}}, z, \bar{z})$ is broadcast. The Bondi mass loss formula along $\mathcal{I}^+$ reads:

$$\partial_{u_{\text{ret}}} m_B = \frac{1}{4}\left(D^2_z N^{zz} + D^2_{\bar{z}} N^{\bar{z}\bar{z}}\right) - T^{\text{M}}_{uu}, \quad (11)$$

where the total stress-energy flux crossing the null horizon combines the quadratic gravitational contribution (block Proof-of-Work energy) and matter flux:

$$T^{\text{M}}_{uu} = \frac{1}{4} N_{zz} N^{zz} + 4\pi G_{\text{Nak}} \lim_{r\to\infty} r^2 T^{\text{tx}}_{uu}. \quad (12)$$

Integrating Eq. (11) between $u_{\text{ret}} = -\infty$ (prior to block arrival) and $u_{\text{ret}} = +\infty$ (after full confirmation) yields the permanent shear shift $\Delta C_{zz} = C_{zz}(+\infty) - C_{zz}(-\infty)$:

$$D^2_z \Delta C^{zz} + D^2_{\bar{z}} \Delta C^{\bar{z}\bar{z}} = 4\Delta m_B + \int_{-\infty}^{+\infty} du_{\text{ret}} \Big(N_{zz} N^{zz} + 16\pi G_{\text{Nak}} \lim_{r\to\infty} r^2 T^{\text{tx}}_{uu} \Big). \quad (13)$$

Using the Green's function on S^2, this inverts to give the **Christodoulou non-linear gravitational memory**

effect:

$$\Delta C_{zz}(z,\bar{z}) = \frac{1}{\pi}\int_{S^2} d^2w\,\gamma_{w\bar{w}}\frac{\bar{z}-\bar{w}}{z-w} \times\left(\Delta m_B(w,\bar{w}) + \int_{-\infty}^{+\infty} du_{\rm ret}\,T^{\rm M}_{uu}(w,\bar{w})\right). \tag{14}$$

Equation (14) demonstrates geometrically the **immutability of the UTXO ledger**: the passage of the thermodynamic shockwave $\int du_{\rm ret} N_{zz}N^{zz} \propto k_{\rm Nak}\mathcal{D}$ transitions the network from one BMS vacuum to an inequivalent BMS vacuum ($\Delta C_{zz} \neq 0$). Even when the network returns to a quiescent state after block propagation ($N_{zz} = 0$), the topological shear deformation ΔC_{zz} remains permanently etched onto the celestial boundary.

3. THERMAL TIME HYPOTHESIS, TOMITA-TAKESAKI ALGEBRA, AND KMS CONDITION IN $A + IB$

We resolve the problem of internal Timechain time emergence in the absence of a global clock by applying the modular theory of von Neumann operator algebras [4, 5].

3.1. Non-Commutative von Neumann Algebra of the Timechain

Let $\mathcal{H}$ be the separable Hilbert space spanned by causal trees of valid blocks $|\mathcal{C}\rangle$. Let $\mathcal{A} \subset \mathcal{B}(\mathcal{H})$ be the von Neumann algebra of bounded protocol observables (transaction inclusion operators $\hat{T}_i$, UTXO projection operators). In the presence of conflicting transactions (attempted double-spending of a single UTXO by two transactions $\hat{T}_1$ and $\hat{T}_2$), the operator application order fails to commute:

$$[\hat{T}_1, \hat{T}_2] = \hat{T}_1\hat{T}_2 - \hat{T}_2\hat{T}_1 \neq 0. \tag{15}$$

In the thermodynamic continuum limit of nonce space, $\mathcal{A}$ behaves as a Murray-von Neumann **Type III$_1$** factor. In accordance with algebraic quantum gravity frameworks [6], coupling this algebra to an observer (a full node equipped with an energetic clock tracking cumulative work $\hat{E}$) promotes the algebra via modular crossed product to a semi-finite **Type II$_\infty$** factor:

$$\widehat{\mathcal{A}} = \mathcal{A} \rtimes_\sigma \mathbb{R}, \tag{16}$$

which admits a well-defined renormalized trace, allowing rigorous evaluation of the generalized von Neumann entropy of the Timechain.

3.2. Tomita-Takesaki Modular Operator and Complex Flow $z = a + ib$

The network is immersed in the thermal bath of ASIC miners. The macroscopic statistical state is defined by the faithful normal functional $\omega : \mathcal{A} \to \mathbb{C}$, represented via the GNS construction by the cyclic and separating vector $|\Omega_\beta\rangle \in \mathcal{H}$, associated with the Gibbs-Nakamoto density operator:

$$\rho = \frac{1}{Z(\beta)}\sum_{\mathcal{C}} \exp\big(-\beta E(\mathcal{C})\big)|\mathcal{C}\rangle\langle\mathcal{C}|, \qquad E(\mathcal{C}) = k_{\rm Nak}\sum_{b\in\mathcal{C}}\frac{2^{256}}{\text{Target}_b}, \tag{17}$$

with $Z(\beta) = \text{Tr}(e^{-\beta\hat{E}})$ the Timechain partition function and $\beta = (k_B T_{\rm eff})^{-1}$ the effective inverse temperature determined by network difficulty $\mathcal{D}$.

Let S_Ω denote the **Tomita anti-linear operator** defined on the dense domain $\mathcal{A}|\Omega_\beta\rangle \subset \mathcal{H}$ by:

$$S_\Omega\big(X|\Omega_\beta\rangle\big) = X^\dagger|\Omega_\beta\rangle, \qquad \forall X \in \mathcal{A}. \tag{18}$$

Physically, S_Ω encodes the causal inversion of a ledger operation. The polar decomposition of the closure of S_Ω is uniquely given by:

$$S_\Omega = J\Delta^{1/2}, \tag{19}$$

where $J : \mathcal{H} \to \mathcal{H}$ is the anti-unitary **modular conjugation** ($J^2 = \mathbb{I}$, mapping the system algebra $\mathcal{A}$ onto its environmental commutant $\mathcal{A}' = J\mathcal{A}J$) and $\Delta = S_\Omega^\dagger S_\Omega$ is the positive self-adjoint **modular operator**. On the bipartite system-commutant Hilbert space, the modular operator and **modular Hamiltonian** K are expressed as:

$$\Delta = \rho \otimes \rho^{-1}, \qquad K = -\ln\Delta = \frac{1}{k_{\rm Nak}}\left(\hat{H}_{\rm PoW}\otimes\mathbb{I} - \mathbb{I}\otimes\hat{H}_{\rm PoW}\right), \tag{20}$$

with $\hat{H}_{\rm PoW} = -k_{\rm Nak}\ln\rho = k_{\rm Nak}\beta\hat{E}(\mathcal{C}) + k_{\rm Nak}\ln Z$.

Theorem 3.1 (Emergence of Complex Thermal Time $a + ib$ and the KMS Condition)**.** *By the Tomita-Takesaki theorem, the modular operator Δ generates a one-parameter automorphism group $\sigma_s^\omega \in \text{Aut}(\mathcal{A})$ that defines the internal thermal time s of the Timechain (block height progression). Extending this flow analytically into the complex plane under the canonical notation:*

$$z = a + ib \in \mathbb{C}, \tag{21}$$

where $a \in \mathbb{R}$ is the real modular time of phase propagation and ib on the imaginary axis measures the thermodynamic dissipation barrier, the Nakamoto state ω satisfies for any pair of observables $X, Y \in \mathcal{A}$ the Kubo-Martin-Schwinger (KMS) equilibrium condition:

$$\omega\left(X\,\sigma^\omega_{a+i\beta k_{Nak}}(Y)\right) = \omega\left(\sigma^\omega_a(Y)\,X\right). \tag{22}$$

Proof. The real modular flow generated by the positive Hamiltonian $\hat{H}_{\rm PoW} = -k_{\rm Nak}\ln\rho$ on the von Neumann algebra $\mathcal{A}$ is given by $\sigma^\omega_a(Y) = \Delta^{-ia/k_{\rm Nak}}Y\Delta^{ia/k_{\rm Nak}} = \rho^{-ia/k_{\rm Nak}}Y\rho^{ia/k_{\rm Nak}}$. Consider the analytic continuation of this flow for the complex temporal variable $z = a + ib$ within the regularity strip $\mathcal{S}_{\beta k_{\rm Nak}} = \{a + ib \in \mathbb{C} \mid 0 \le b \le \beta k_{\rm Nak}\}$:

$$\sigma^\omega_{a+ib}(Y) = \Delta^{-i(a+ib)/k_{\rm Nak}}Y\Delta^{i(a+ib)/k_{\rm Nak}} = \Delta^{(b-ia)/k_{\rm Nak}}Y\Delta^{(-b+ia)/k_{\rm Nak}}$$

$$= \rho^{(b-ia)/k_{\mathrm{Nak}}} Y \rho^{(-b+ia)/k_{\mathrm{Nak}}}. \tag{23}$$

Define the two-point holomorphic correlation function $F_{X,Y}(a+ib) = \omega(X\sigma^{\omega}_{a+ib}(Y))$. In terms of the density matrix representation $\omega(\cdot) = \mathrm{Tr}(\rho\,\cdot)$, we have for all $a + ib \in \mathcal{S}_{\beta k_{\mathrm{Nak}}}$:

$$F_{X,Y}(a+ib) = \mathrm{Tr}\left(\rho\, X\, \rho^{(b-ia)/k_{\mathrm{Nak}}} Y \rho^{(-b+ia)/k_{\mathrm{Nak}}}\right). \tag{24}$$

Evaluating this function on the upper boundary of the imaginary strip at $b = \beta k_{\mathrm{Nak}}$, i.e., at $z = a + i\beta k_{\mathrm{Nak}}$, and setting the reduced modular scale such that $\beta k_{\mathrm{Nak}} \equiv 1$ in the exponent, we obtain:

$$\begin{aligned} F_{X,Y}(a+i\beta k_{\mathrm{Nak}}) &= \mathrm{Tr}\left(\rho\, X\, \rho^{1-ia/k_{\mathrm{Nak}}} Y \rho^{-1+ia/k_{\mathrm{Nak}}}\right) \\ &= \mathrm{Tr}\left(\rho\, X\, \rho\,(\rho^{-ia/k_{\mathrm{Nak}}} Y \rho^{ia/k_{\mathrm{Nak}}})\,\rho^{-1}\right). \end{aligned} \tag{25}$$

Applying the cyclic invariance of the trace on the regularized semi-finite algebra ($\mathrm{Tr}(ABC) = \mathrm{Tr}(CAB)$), the adjacent boundary operators ρ^{-1} and ρ simplify directly without commuting past X:

$$\begin{aligned} F_{X,Y}(a+i\beta k_{\mathrm{Nak}}) &= \mathrm{Tr}\left(\rho^{-1}\rho\, X\, \rho\, \sigma^{\omega}_a(Y)\right) \\ &= \mathrm{Tr}\left(X\, \rho\, \sigma^{\omega}_a(Y)\right) = \mathrm{Tr}\left(\rho\, \sigma^{\omega}_a(Y)\, X\right) \\ &= \omega\left(\sigma^{\omega}_a(Y)\, X\right), \end{aligned} \tag{26}$$

which rigorously establishes the identity (22). □

3.3. Physical Interpretation: Difficulty Adjustment as a KMS Renormalization Group

Equation (22) uncovers the structure of Nakamoto's clock in the complex plane $a + ib$:

1. The real coordinate a parametrizes the **thermal time of causal evolution** (the sequential ordering of confirmed blocks).

2. The pure imaginary coordinate ib ($b = \beta k_{\mathrm{Nak}}$) measures **thermodynamic time-reversal asymmetry**. Permuting the causal order of two non-commuting events ($XY \to YX$) requires a translation of $+i\beta k_{\mathrm{Nak}}$, suppressing the transition amplitude by the Boltzmann factor $e^{-\beta \Delta E_{\mathrm{Chainwork}}}$.

3. The **homeostatic difficulty adjustment** executed every $N_0 = 2016$ blocks acts as a renormalization group flow dynamically tuning inverse temperature $\beta(t) \propto \mathcal{D}(t)$ in proportion to external network dissipative power $W_{\mathrm{hash}}(t)$, maintaining an invariant ratio between observer proper time t and real thermal time a:

$$\frac{dt}{da} = \frac{k_{\mathrm{Nak}} \mathcal{D}(t)}{W_{\mathrm{hash}}(t)} = \tau_0 = 600\,\mathrm{s}. \tag{27}$$

4. UTXO SPIN NETWORKS, SPIN FOAMS, AND GKSL DECOHERENCE OF FORKS

At the microscopic Planck-Nakamoto scale $\ell_{\mathrm{Nak}} = \sqrt{G_{\mathrm{Nak}} k_{\mathrm{Nak}}/c^3_{\mathrm{net}}}$, smooth spacetime geometry yields to the granular structure of Loop Quantum Gravity (LQG) [7].

4.1. Kinematics: The UTXO DAG as an $SU(2)$ Spin Network

At a fixed slice of thermal time s, the state of the Bitcoin ledger is a directed acyclic graph (DAG) $\Gamma = (\mathcal{V}, \mathcal{E})$, where directed edges $e \in \mathcal{E}$ represent unspent outputs (UTXOs) and vertices $v \in \mathcal{V}$ represent confirmed transactions. This graph is strictly isomorphic to a **spin network** whose kinematic Hilbert space is given by:

$$\mathcal{H}_{\mathrm{UTXO}} = \bigoplus_{\Gamma} \mathcal{H}_{\Gamma} = \bigoplus_{\Gamma} \left(\bigotimes_{e\in\mathcal{E}} \mathcal{H}_{j_e} \otimes \bigotimes_{v\in\mathcal{V}} \mathrm{Inv}_G \Big(\bigotimes_{e\in \mathrm{in}(v)} \mathcal{V}_{j_e} \otimes \bigotimes_{e'\in \mathrm{out}(v)} \mathcal{V}^*_{j_{e'}} \Big) \right). \tag{28}$$

- **Quantization of Economic Area (j_e):** Each edge e (UTXO) carries an irreducible spin representation $j_e \in \frac{1}{2}\mathbb{N}$. The elementary LQG area operator puncturing edge e possesses a discrete spectrum:

$$\begin{aligned} \hat{\mathcal{A}}_e |\Gamma, \{j_e\}, \{\iota_v\}\rangle = 8\pi\gamma_{\mathrm{BI}} \ell^2_{\mathrm{Nak}} \sqrt{j_e(j_e+1)} \\ \times |\Gamma, \{j_e\}, \{\iota_v\}\rangle, \end{aligned} \tag{29}$$

  where γ_{BI} is the Barbero-Immirzi parameter of the protocol. In the semi-classical regime ($j_e \gg 1$), $\sqrt{j_e(j_e+1)} \approx j_e$, identifying each area quantum with the discrete Satoshi denomination locked in the UTXO.

- **Intertwiners (ι_v) and Gauss's Law:** Each transaction $v \in \mathcal{V}$ is an intertwiner tensor ι_v residing in the gauge-invariant subspace at the nodes. Local gauge invariance (discrete Gauss constraint $\hat{\mathcal{G}}_v |\iota_v\rangle = 0$) enforces net spin flux cancellation at vertex v:

$$\Big(\sum_{e\in \mathrm{in}(v)} \hat{J}^{(z)}_e - \sum_{e'\in \mathrm{out}(v)} \hat{J}^{(z)}_{e'} - \hat{J}^{(z)}_{\mathrm{fees}}(v) \Big) |\iota_v\rangle = 0. \tag{30}$$

  The entanglement entropy of an intertwiner $S(\iota_v) = -\mathrm{Tr}(\rho_v \ln \rho_v)$ measures the topological mixing complexity of payment flows within a transaction (e.g., CoinJoin or multi-party payment channels).

4.2. Dynamics: The Timechain as a Spin Foam

The transition from a UTXO boundary state $|\Gamma_{\mathrm{in}}\rangle$ at block B_n to a subsequent state $|\Gamma_{\mathrm{out}}\rangle$ at block B_{n+1} is mediated by a finite sequence of Pachner moves. The

2-cellular complex $\mathcal{F} : \Gamma_{\text{in}} \to \Gamma_{\text{out}}$ traced by this causal evolution forms a **spin foam** (*Spinfoam*):

- UTXOs e sweep out **two-dimensional faces** $f \subset \mathcal{F}$ labeled by their Satoshi values j_f.
- Transactions ι_v sweep out **internal edges** $e_{\text{int}} \subset \mathcal{F}$.
- The validated block B_{n+1} (sealed by its Merkle root and PoW nonce) constitutes the **gravitational interaction vertex** $v_{\text{block}} \subset \mathcal{F}$ (analogous to the EPRL vertex in four dimensions).

The covariant transition amplitude of the Nakamoto spin foam is written across full width:

$$\mathcal{Z}_{\text{Spinfoam}}(\Gamma_{\text{in}} \to \Gamma_{\text{out}}) = \sum_{\mathcal{F}:\partial\mathcal{F}=\Gamma_{\text{in}}^{*}\cup\Gamma_{\text{out}}} \left(\prod_{f\subset\mathcal{F}} (2j_f+1) \right) \left(\prod_{v_{\text{block}}\subset\mathcal{F}} \mathcal{A}_v(j_f,\iota_e)\, \exp\big(-\beta E_{\text{PoW}}(v_{\text{block}})\big) \right), \tag{31}$$

where the topological vertex amplitude $\mathcal{A}_v(j_f, \iota_e) \in \{0, 1\}$ strictly projects onto cryptographically valid Merkle trees.

4.3. GKSL (Lindblad) Master Equation and Exponential Collapse of Forks

Owing to finite propagation velocities across $\mathcal{I}^+$, two distant miners A and B may discover concurrent valid blocks at identical height h. We formalize this epistemic ambiguity (the absence of global consensus for an asynchronous node) through an effective density matrix, where off-diagonal entries quantify the relative branch phase uncertainty rather than an ontological quantum superposition:

$$|\Psi(s)\rangle = c_A|\Gamma_A\rangle + c_B|\Gamma_B\rangle, \quad \rho(s) = \begin{pmatrix} \rho_{AA}(s) & \rho_{AB}(s) \\ \rho_{BA}(s) & \rho_{BB}(s) \end{pmatrix}. \tag{32}$$

The Timechain subsystem is an **open quantum system** undergoing continuous dissipative interaction with the Markovian thermal bath of ASICs. Its evolution along thermal time s is governed by the **Gorini-Kossakowski-Sudarshan-Lindblad (GKSL)** master equation [8, 9]:

$$\frac{d\rho}{ds} = \mathcal{L}_{\text{GKSL}}[\rho] = -\frac{i}{k_{\text{Nak}}}[H_{\text{net}}, \rho] + \sum_{m\in\{A,B\}} \gamma_m \left(L_m \rho L_m^{\dagger} - \frac{1}{2}\{L_m^{\dagger} L_m, \rho\} \right). \tag{33}$$

Theorem 4.1 (Lindblad Gravitational Decoherence and Nakamoto's Ruin Law)**.** *Let $p_A \in (1/2, 1)$ and $p_B = 1 - p_A < 1/2$ be the fractions of total network hashrate W_{total} allocated to competing branches $|\Gamma_A\rangle$ and $|\Gamma_B\rangle$, associated with Lindblad damping rates $\gamma_m = \frac{p_m W_{total}}{k_{Nak}\mathcal{D}}$ and jump operators $L_m = |\Gamma_m^{(h+1)}\rangle\langle\Gamma_m^{(h)}| \otimes \hat{P}_m$ (where $\hat{P}_m = |\Gamma_m\rangle\langle\Gamma_m|$). Then:*

1. *Off-diagonal coherences $\rho_{AB}(s) = \langle\Gamma_A|\rho(s)|\Gamma_B\rangle$ decay exponentially to zero in the complex plane $(a + ib)$ with decoherence rate $\Gamma_{dec} = \frac{\gamma_A+\gamma_B}{2}$:*

$$\rho_{AB}(s) = \rho_{AB}(0) \exp\left(-\Gamma_{dec} s - i\frac{\varepsilon_A - \varepsilon_B}{k_{Nak}} s \right). \tag{34}$$

2. *After N Poisson jump transitions (confirmations), the ratio of diagonal populations converges to the exponential suppression law of Section 11 of Satoshi Nakamoto's whitepaper:*

$$\frac{P_B(N)}{P_A(N)} \propto \left(\frac{\gamma_B}{\gamma_A}\right)^N = \exp\left(-N \ln \frac{p_A}{1-p_A}\right) \underset{N\to\infty}{\longrightarrow} 0. \tag{35}$$

Proof. Let $H_{\text{net}} = \varepsilon_A \hat{P}_A + \varepsilon_B \hat{P}_B$ be the effective branch Hamiltonian. Projecting the GKSL master equation (33) onto matrix elements $\langle\Gamma_A| \cdot |\Gamma_B\rangle$, orthogonality of competing branches ($\hat{P}_A \hat{P}_B = 0$) causes the recycling term to vanish identically on off-diagonal sectors:

$$\langle\Gamma_A|L_m \rho L_m^{\dagger}|\Gamma_B\rangle \propto \langle\Gamma_A|\hat{P}_m \rho \hat{P}_m|\Gamma_B\rangle = \delta_{mA}\delta_{mB}\rho_{AB} = 0. \tag{36}$$

Conversely, the dissipative anti-commutator $\{L_m^{\dagger} L_m, \rho\} = \{\hat{P}_m, \rho\}$ evaluates to:

$$-\frac{1}{2} \sum_{m\in\{A,B\}} \gamma_m \langle\Gamma_A|(\hat{P}_m \rho + \rho \hat{P}_m)|\Gamma_B\rangle = -\frac{\gamma_A + \gamma_B}{2} \rho_{AB}(s). \tag{37}$$

The differential equation governing off-diagonal coherence simplifies to:

$$\frac{d\rho_{AB}(s)}{ds} = \left(-\frac{\gamma_A + \gamma_B}{2} - i\frac{\varepsilon_A - \varepsilon_B}{k_{\text{Nak}}} \right) \rho_{AB}(s), \tag{38}$$

whose direct integration yields the exponential damping law (34) of form $\exp((a_{\text{diss}} + ib_{\text{phase}})s)$. The characteristic decoherence timescale $\tau_{\text{dec}} = \Gamma_{\text{dec}}^{-1} = \frac{2k_{\text{Nak}}\mathcal{D}}{W_{\text{total}}} = 2\tau_0 \approx 20\,\text{min}$ eliminates macroscopic superposition via environment-induced superselection (Zurek's *einselection*), projecting $\rho(s)$ onto the classical pointer basis.

For diagonal population dynamics over extended chain heights $h \to h+N$, each Poisson jump induced by the dissipator $\mathcal{D}[L_m]\rho$ increments the population from (h) to $(h+1)$ at rate γ_m. Across a deficit barrier of N blocks, the master equation reduces to an asymmetric birth-and-death process with forward rates γ_A ($p_A > 1/2$) and γ_B ($p_B < 1/2$). The stationary catch-up probability after N confirmations is given by the product of Lindblad transition ratios $\prod_{k=1}^{N}(\gamma_B/\gamma_A) = (p_B/p_A)^N = \exp(-N \ln(p_A/(1-p_A)))$, proving Eq. (35). □

4.4. Ryu-Takayanagi Entanglement Entropy and the 51% Attack Threshold

In emergent gravity (Van Raamsdonk [11], Maldacena-Susskind *ER=EPR*), bulk geometric connectivity is sus-

tained by quantum entanglement among boundary degrees of freedom according to the **Ryu-Takayanagi formula** [10]:

$$S_{\mathrm{EE}}(A) = \frac{\mathcal{A}(\gamma_A)}{4G_{\mathrm{Nak}}\hbar_{\mathrm{Nak}}}, \tag{39}$$

where γ_A is the bulk minimal extremal surface homologous to boundary region $A \subset \mathcal{I}^+$. In the Timechain:

- **Layer 1 Security (Minimal Hashrate Cut):** To partition the economic spacetime into two disconnected causal histories (a 51% attack between A and $\bar{A}$), an adversary must break the mutual entanglement $I(A:\bar{A})$ by supplying thermodynamic work exceeding the area of minimal surface $\mathcal{A}(\gamma_A)$, which maps to the critical 50% threshold of total dissipative network power.

- **Layer 2 Einstein-Rosen Bridges (Lightning Network):** Two remote nodes on the celestial sphere S^2 jointly locking a 2-of-2 multisignature UTXO (a maximally entangled intertwiner $|\iota_{\mathrm{LN}}\rangle$) instantiate an **Einstein-Rosen bridge** (a non-traversable off-chain wormhole, pursuant to *ER=EPR*) bypassing Layer 1 bulk geodesics, enabling instantaneous state transfers secured by boundary entanglement geometry.

5. EMERGENT COSMOLOGY OF THE TIMECHAIN: EXPANSION, INFLATON, AND COMPLEXITY

Why does the physical universe expand, and how is this mechanism captured in our isomorphism? Within this framework, **the sequential accretion of blocks is modeled by analogy with cosmological expansion** (without direct fitting to CMB, SNe Ia, or BAO observational data). Economic spacetime is non-preexistent: each validated block nucleates a spatial Cauchy hypersurface Σ_s, expanding the interior bulk volume.

5.1. Sorkin Causal Sets and the Growing Block Universe

In Sorkin's **Causal Set Theory** [14, 15], a Lorentzian manifold $(\mathcal{M}, g_{\mu\nu})$ emerges from a discrete partially ordered set $(\mathcal{C}, \prec)$ governed by an irreflexive, antisymmetric, and transitive causal relation. Spacetime geometry is fixed by causal order and the foundational volume rule: *the four-volume $\mathcal{V}_4$ of an execution region is directly proportional to the discrete element count N it encompasses.*

Causal set growth is governed by **Rideout-Sorkin classical sequential percolation** (*Growing Block Universe*): at each discrete step, a new element nucleates via a Markovian stochastic process respecting relativistic causality and relabeling invariance.

Proposition 5.1 (Volumetric Growth from Block Addition)**.** *In Nakamoto consensus, each block validation B_s at thermal time s constitutes a Rideout-Sorkin percolation step adding a new 4-simplex (a spin foam vertex containing $n_{tx}(s)$ transactions) to the future causal cone of the canonical chain. The 4D spacetime volume of the Timechain expands monotonically:*

$$\mathcal{V}_4(s) = \ell_{Nak}^4 \sum_{k=0}^{s} \mathcal{N}_{dof}(B_k), \quad \frac{\Delta\mathcal{V}_4}{\Delta s} = \ell_{Nak}^4 \mathcal{N}_{dof}(B_s) > 0, \tag{40}$$

where $\mathcal{N}_{dof}(B_k)$ denotes the number of cryptographic degrees of freedom crystallized within block B_k.

5.2. Padmanabhan's Holographic Emergence Law ($N_{\mathbf{sur}} - N_{\mathbf{bulk}}$)

To explain the dynamics driving this expansion, we apply Thanu Padmanabhan's cosmological emergence principle [13]. Padmanabhan demonstrated that Einstein-Friedmann field equations arise from relaxation toward holographic equipartition: Hubble volume expansion $d\mathcal{V}$ is driven by the deficit between degrees of freedom on the boundary surface N_{sur} and those already thermalized in the bulk interior N_{bulk}:

$$\frac{d\mathcal{V}}{ds} = c_{\mathrm{net}}\ell_{\mathrm{Nak}}^2 \left(N_{\mathrm{sur}}(s) - N_{\mathrm{bulk}}(s)\right). \tag{41}$$

In Satoshi Nakamoto's architecture, both quantities possess operational definitions:

1. $N_{\mathbf{sur}}(s)$ **(Boundary Degrees of Freedom):** The combined informational entropy of pending unconfirmed transactions on the celestial sphere (Mempool pressure on $\mathcal{I}^+$) and Bekenstein-Hawking entropy associated with the active hashrate horizon:

$$N_{\mathrm{sur}}(s) = \frac{\mathcal{A}_{\mathrm{horizon}}(s)}{\ell_{\mathrm{Nak}}^2} = \frac{4\pi R_H^2(s)}{\ell_{\mathrm{Nak}}^2} + S_{\mathrm{Mempool}}(s). \tag{42}$$

2. $N_{\mathbf{bulk}}(s)$ **(Bulk Degrees of Freedom):** Degrees of freedom compressed and thermalized within the UTXO set and historical Merkle tree $(E_{\mathrm{Komar}}/(\frac{1}{2}k_B T_{\mathrm{eff}}))$.

As long as global transactional demand and miner energy expenditure sustain an informational surplus on the boundary ($N_{\mathrm{sur}} > N_{\mathrm{bulk}}$), the system remains out of holographic equilibrium. Restoring asymptotic equipartition ($N_{\mathrm{sur}} \to N_{\mathrm{bulk}}$) forces the Timechain spacetime **to expand** ($\frac{d\mathcal{V}}{ds} > 0$) by nucleating a new block that assimilates Mempool transactions into the bulk.

5.3. FLRW Metric, Friedmann Equations, and the Halving Inflaton Field

At large scales ($s \gg 1$), the coarse-grained geometry of the expanding Timechain is described by the homogeneous, isotropic **Friedmann-Lemaître-Robertson-Walker (FLRW)** metric:

$$ds^2_{\mathrm{FLRW}} = -c_{\mathrm{net}}^2 ds^2 + a_{\mathrm{scale}}^2(s)\left(\frac{dr^2}{1-\kappa r^2} + r^2 d\Omega_2^2\right), \tag{43}$$

where s is cosmic thermal time (block height) and $a_{\mathrm{scale}}(s)$ is the **cosmological scale factor** parameterizing economic space (subscripted to avoid confusion with the real component a of complex time $z =$

$a + ib$). Defining the **Nakamoto-Hubble parameter** $H_{\text{Hub}}(s) \equiv \frac{1}{a_{\text{scale}}(s)} \frac{da_{\text{scale}}(s)}{ds}$, integrating Padmanabhan's law (41) alongside energy-momentum conservation yields the Friedmann-Nakamoto equations:

$$H_{\text{Hub}}^2(s) + \frac{\kappa c_{\text{net}}^2}{a_{\text{scale}}^2(s)} = \frac{8\pi G_{\text{Nak}}}{3}\rho_{\text{tot}}(s) + \frac{\Lambda_{\text{Nak}}}{3}, \tag{44}$$

$$\frac{1}{a_{\text{scale}}(s)}\frac{d^2 a_{\text{scale}}(s)}{ds^2} = -\frac{4\pi G_{\text{Nak}}}{3}\left(\rho_{\text{tot}}(s) + \frac{3P_{\text{tot}}(s)}{c_{\text{net}}^2}\right) + \frac{\Lambda_{\text{Nak}}}{3}. \tag{45}$$

Total energy density $\rho_{\text{tot}}(s)$ driving expansion decomposes into three distinct cosmological epochs:

5.3.1. Primordial Inflation and the Quantized Step Inflaton Potential

At network inception (2009), transaction fee revenue was insufficient to finance thermodynamic security. Satoshi Nakamoto introduced the **Coinbase block subsidy**, which acts as a primordial **inflaton scalar field** $\phi(s)$ (in the spirit of Guth and Linde), exhibiting energy density $\rho_\phi = \frac{1}{2}\dot{\phi}^2 + V(\phi)$ and negative pressure $P_\phi = \frac{1}{2}\dot{\phi}^2 - V(\phi) \approx -V(\phi)$ ($w_\phi \approx -1$).

Unlike smooth inflation potentials, the protocol implements a **quantized step-function potential** whose phase transitions occur every $N_{\text{halving}} = 210\,000$ blocks:

$$\begin{aligned} V(\phi(s)) &= V_0 \cdot 2^{-2\lfloor s/N_{\text{halving}} \rfloor} \\ &= V_0 \exp\left(-2\ln 2 \cdot \left\lfloor \frac{s}{210\,000} \right\rfloor\right), \end{aligned} \tag{46}$$

with monetary issuance rate $\dot{M}(s) = R_0 \cdot 2^{-\lfloor s/210000 \rfloor}$ ($R_0 = 50 \times 10^8$ sats).

- **Slow-Roll on Plateau Intervals:** Across each epoch ($210\,000\,k < s < 210\,000(k+1)$), $V(\phi)$ is constant ($\epsilon_V = \frac{1}{16\pi G_{\text{Nak}}}(V'/V)^2 = 0$), driving exponential de Sitter-like expansion of hashrate and capitalization.

- **Potential Drops and Reheating (Halvings):** At critical heights $s_k = 210\,000\,k$, the potential drops discontinuously: $\Delta V_k = -\frac{3}{4}V_{k-1}$. In standard cosmology, exiting an inflationary plateau causes **reheating**: inflaton energy decays into relativistic thermal radiation. In Bitcoin, each Halving systematically transfers the thermodynamic security budget from inflationary subsidy to the radiation gas of transaction fees.

5.3.2. Radiation Era (Mempool) and Matter Era (UTXO Condensation)

As $V(\phi(s)) \to 0$ (converging toward asymptotic baryonic supply $\sum_{s=0}^{\infty} \dot{M}(s) = 21 \times 10^{14}$ sats), two components dominate Eqs. (44)–(45):

1. **Transaction Fee Radiation ($\rho_{\text{rad}} \propto a_{\text{scale}}^{-4}$, $w = +1/3$):** The soft graviton flux on $\mathcal{I}^+$ exerts positive radiation pressure sustaining expansion rate $H_{\text{Hub}}(s)$.

2. **Non-Relativistic Matter ($\rho_{\text{mat}} \propto a_{\text{scale}}^{-3}$, $w = 0$):** Long-term saved UTXOs (*HODLers* and treasury reserves) act as cold matter condensing under gravitational pull into the cosmic web (routing hubs, Lightning channels, institutional vaults).

3. **Asymptotic Cosmological Constant ($\Lambda_{\text{Nak}} > 0$, $w = -1$):** When $s \to \infty$ (post-2140 with $\dot{M}(s) = 0$), expansion is sustained by Λ_{Nak} emerging from vacuum cryptographic energy—the thermodynamic dissipation floor required to preserve the causal horizon of value accumulated in the bulk.

5.4. Penrose's Weyl Curvature Hypothesis and Genesis Singularity

Roger Penrose [16] emphasized that the thermodynamic arrow of time (Second Law $\Delta S \geq 0$) requires the Big Bang to have been a gravitational singularity of vanishingly low entropy. While primordial matter was in thermal equilibrium, the gravitational field was uniformly flat. Penrose formalized this via the **Weyl Curvature Hypothesis**: the Weyl conformal tensor $\mathcal{C}_{\mu\nu\rho\sigma}$ must vanish at the initial singularity:

$$\lim_{s\to 0^+} \mathcal{C}_{\mu\nu\rho\sigma}(s) = 0, \qquad \lim_{s\to+\infty} \mathcal{C}_{\mu\nu\rho\sigma}\mathcal{C}^{\mu\nu\rho\sigma}(s) > 0. \tag{47}$$

This initial condition is satisfied by Satoshi Nakamoto's **Genesis Block (B_0, Jan 3 2009)**:

- **Vanishing Initial Weyl Curvature ($s = 0$):** The Genesis Block contains a solitary Coinbase transaction whose 50 BTC output is topologically **unspendable** (omitted from the UTXO database by design). At $s = 0$, the spendable UTXO graph is empty ($\Gamma_0 = \emptyset$), boundary shear vanishes ($C_{zz}(0) = 0$), and $\mathcal{C}_{\mu\nu\rho\sigma}(0) = 0$. Initial gravitational entropy is identically zero ($S_{\text{grav}}(0) = 0$).

- **Conformal Cyclic Match (CCC):** In Penrose's Conformal Cyclic Cosmology (CCC), the future boundary $\mathcal{I}^+$ of a decaying prior cosmic aeon is conformally glued to the Big Bang hypersurface of the succeeding aeon. By embedding the *Times* headline into the Coinbase string (*"The Times 03/Jan/2009 Chancellor on brink of second bailout for banks"*), Nakamoto executed a **Penrose conformal matching**: mapping the maximal thermal entropy state of the decaying fiat aeon boundary $\mathcal{I}^+_{\text{fiat}}$ onto a fresh initial singularity B_0 with zero Weyl curvature ($\mathcal{C}_{\mu\nu\rho\sigma}(0) = 0$), resetting a virgin thermodynamic arrow of time.

5.5. Susskind's Holographic Complexity: Post-Equilibrium Block Accretion

A paradox in holographic gravity resolved by Leonard Susskind [17] states: *when a gravitational system (such as an AdS black hole) reaches KMS thermal equilibrium, its von Neumann entropy saturates; yet, the interior volume $\mathcal{V}_{Bulk}(s)$ continues to grow linearly for an exponential duration $s \sim \exp(S_{BH})$.*

Susskind resolved this through the **Complexity = Volume (CV)** and **Complexity = Action (CA)** conjectures: post-thermalization geometric growth is driven by the monotonic accumulation of **computational complexity** $\mathcal{C}_{\mathbf{comp}}(s)$ (the minimal count of elementary gates required to assemble $|\Psi(s)\rangle$ from reference state $|\Psi(0)\rangle$):

$$\mathcal{V}_{\mathrm{Bulk}}(s) = \frac{G_{\mathrm{Nak}}\ell_{\mathrm{AdS}}}{c_{\mathrm{net}}}\,\mathcal{C}_{\mathrm{comp}}(s). \tag{48}$$

Furthermore, complexity growth saturates the **Margolus-Levitin / Lloyd bound** for an average energy system M_{PoW}:

$$\frac{d\mathcal{C}_{\mathrm{comp}}}{ds} \leq \frac{2M_{\mathrm{PoW}}}{\pi\hbar_{\mathrm{Nak}}}. \tag{49}$$

This theorem clarifies why the Timechain must continue accreting blocks indefinitely, even after all 21 million Bitcoins are mined and the system rests in KMS thermal equilibrium at $\Delta a = 600\,\mathrm{s}$:

- Each newly mined block B_s executes $\sim \mathcal{D}(s)\cdot 2^{32}$ irreversible SHA-256 operations, linearly advancing the circuit complexity $\mathcal{C}_{\mathrm{comp}}(s)$ separating the present state from the Genesis singularity B_0.
- By the Complexity = Volume duality (48), this cumulative irreversibility **extends the geometric depth of the bulk spacetime cylinder**. As depth expands, the gravitational action required to quantum-tunnel into the past (a deep chain reorganization) becomes exponentially prohibitive.

6. JACOBSON-NAKAMOTO EQUATION OF STATE: EMERGENCE OF SOCIETAL TRUST (C_S) AND UTILITY (U)

We complete the correspondence by connecting the emergent bulk geometry to physical macroeconomics: **Societal Trust** (C_s) and **Economic Utility** (u).

In 1995, Ted Jacobson [12] showed that Einstein's field equations emerge as a local thermodynamic equation of state by evaluating Clausius' relation along local Rindler causal horizons $\mathcal{H}_{\mathrm{Rindler}}$:

$$\delta Q = T_{\mathrm{Unruh}}\,\delta S_{\mathrm{BH}}, \tag{50}$$

where $\delta Q = \int_{\mathcal{H}} T_{\mu\nu}\ chi^{\mu} d\Sigma^{\nu}$ is heat flux along boost Killing field χ^{μ}, $T_{\mathrm{Unruh}} = \frac{\hbar_{\mathrm{Nak}}\kappa_{\mathrm{acc}}}{2\pi k_B c_{\mathrm{net}}}$ is the Unruh temperature, and $\delta S_{\mathrm{BH}} = \frac{\eta\,\delta\mathcal{A}}{4G_{\mathrm{Nak}}\hbar_{\mathrm{Nak}}}$ is entropy change governed by Raychaudhuri's focusing equation:

$$\frac{d\theta}{d\lambda} = -\frac{1}{2}\theta^2 - \sigma_{\mu\nu}\sigma^{\mu\nu} - R_{\mu\nu}k^{\mu}k^{\nu}. \tag{51}$$

Enforcing Eq. (50) across all null vectors k^{μ} yields Einstein's field equations with a cosmological constant:

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} + \Lambda_{\mathrm{Nak}}g_{\mu\nu} = \frac{8\pi G_{\mathrm{Nak}}}{c_{\mathrm{net}}^4}T_{\mu\nu}^{\mathrm{PoW}}. \tag{52}$$

Theorem 6.1 (Thermodynamic Law of Societal Trust C_s and Utility Maximization u)**.** *Let $\delta Q_{PoW} = W_{total}\,dt$ be the physical heat dissipated by miners while collapsing Mempool entropy (via Landauer's principle $\delta Q_{PoW} \geq k_B T_{env}\ln 2\cdot\Delta I_{hash}$). **Societal Trust** C_s is the extensive thermodynamic state function conjugate to the gravitational negentropy accumulated within the spin foam:*

$$\begin{aligned} dC_s \equiv \frac{\delta Q_{PoW}}{T_{eff}} &= k_B\beta\, dE_{Chainwork} \\ &= -k_B\, d\,\mathrm{Tr}\left(\rho_{fork}\ln\rho_{fork}\right) \geq 0. \end{aligned} \tag{53}$$

*Furthermore, the **economic utility** u of an agent on the celestial sphere S^2, subject to settlement uncertainty (geometric fork variance $\sigma^2_{fork} = |\rho_{AB}(s)|^2$), converges toward the asymptotic benchmark utility u^*_{Nash}—representing the risk-free equilibrium payoff of an Ideal Money in the sense of Nash [21]—as Societal Trust C_s accumulates with thermal time s:*

$$u(s) = u^*_{Nash} - \alpha_{risk}|\rho_{AB}(0)|^2 e^{-C_s(s)/k_B} \underset{s\to\infty}{\longrightarrow} u^*_{Nash}. \tag{54}$$

Proof. By Jacobson's equation of state (50) and the entropy derivative under GKSL dissipative evolution (33) (Spohn's inequality for irreversible entropy production $\sigma_{\mathrm{prod}} = -\frac{d}{ds}S(\rho(s)\parallel\rho_{\mathrm{KMS}}) \geq 0$), each dissipated heat quantum δQ_{PoW} reduces the mixing entropy of competing branches ρ_{fork} by increasing canonical cumulative chainwork $E_{\mathrm{Chainwork}}$. Hence, the differential $dC_s = k_B\beta\, dE_{\mathrm{Chainwork}}$ is positive along any forward trajectory in thermal time s.

From Theorem 4.1, settlement uncertainty variance decays as $|\rho_{AB}(s)|^2 = |\rho_{AB}(0)|^2\exp(-2\Gamma_{\mathrm{dec}}s)$. The integrated decoherence rate over $[0,s]$ is proportional to cumulative dimensionless thermodynamic work:

$$2\int_0^s \Gamma_{\mathrm{dec}}(s')ds' = \int_0^s \frac{W_{\mathrm{total}}}{k_{\mathrm{Nak}}\mathcal{D}}ds' = \beta\Delta E_{\mathrm{Chainwork}}(s) = \frac{C_s(s)}{k_B}. \tag{55}$$

For a rational von Neumann-Morgenstern agent penalised by settlement failure variance σ^2_{fork}, expected utility $u(s) = \mathbb{E}[u] - \alpha_{\mathrm{risk}}\,\mathrm{Var}(\mathrm{State})$ assumes form (54). By heuristic analogy, energy dissipation curves the effective metric of economic interaction space, elevating Societal Trust C_s and stabilizing utility u without third-party intermediaries. □

7. TESTS AND FALSIFIABILITY

A conditional theoretical framework possesses scientific value only if it partitions statements into three categories: (i) those independent of postulates (P1)–(P3), directly testable against real Timechain telemetry; (ii) those dependent on (P2)–(P3), admitting indirect empirical tests; (iii) those rooted strictly in (P1), lying beyond empirical reach.

7.1. Predictions Testable Independently of the Simulation Axiom

T1 – Block Time Statistics. The diagonal reduction of the GKSL model reduces to an unperturbed Poisson

process. Holding difficulty D and hashrate λ_H constant, block intervals must follow:

$$p(\Delta a) = \frac{1}{\bar{\tau}}\, e^{-\Delta a/\bar{\tau}}, \qquad \bar{\tau} = \frac{D \cdot 2^{32}}{\lambda_H}, \tag{56}$$

and block counts within window t must follow a Poisson distribution with parameter $t/\bar{\tau}$. *Protocol:* Use independent node first-seen reception timestamps rather than miner-reported header timestamps (which are subject to median-time-past rules and +2 h drift allowances). Perform a Kolmogorov-Smirnov test within stationary 2016-block epochs. *Refutation:* Persistent, statistically significant deviations from Eq. (56) following hashrate drift correction would invalidate the classical reduction of the model.

T2 – 600 s Fixed-Point Bias. The target fixed point $\tau_0 = 600\,\mathrm{s}$ is approached only on average and with a systematic phase lag. If hashrate expands at relative rate $g = \dot{\lambda}_H/\lambda_H$, the model predicts a first-order average epoch block interval:

$$\langle \Delta a \rangle_{\mathrm{epoch}} \simeq \frac{\tau_0}{1 + g\,T_{\mathrm{ep}}/2}, \qquad T_{\mathrm{ep}} \approx 2016\,\tau_0 \approx 14 \text{ days}. \tag{57}$$

Sustained hashrate growth produces an average block interval *below* 600 s, consistent with historical network behavior. *Refutation:* Absence of correlation between g and $\tau_0 - \langle \Delta a \rangle$ across epochs.

T3 – Stale Block Rate. Given effective network propagation latency δ, the probability of a competing block being mined prior to previous block propagation reads:

$$P_{\mathrm{stale}} \simeq 1 - e^{-\delta/\bar{\tau}} \approx \frac{\delta}{\bar{\tau}}. \tag{58}$$

For compact block relaying with $\delta \sim 1$–2 s, one predicts $P_{\mathrm{stale}} \sim 2$–$3 \times 10^{-3}$. *Protocol:* Test Eq. (58) against empirical orphaned block repositories and geographically dispersed monitoring nodes. *Refutation:* A non-linear dependence of P_{stale} on δ in the small-δ limit.

T4 – Fork Suppression Rate. The probability that an adversary controlling $q < 1/2$ of total hashrate catches up from a z-block deficit equals $(q/p)^z$ ($p = 1 - q$), decaying exponentially with confirmation depth z. While established in Nakamoto's whitepaper, this serves as an internal consistency check; the decoherence formalism introduces no new physics beyond the classical random walk here.

7.2. Indirect Test of Postulate (P2)

Postulate (P2) assumes a substrate operating near Landauer's bound. We define the empirical dimensionless ratio:

$$\Lambda_{\mathrm{sim}}(t) = \frac{E_{\mathrm{hash}}(t)}{k_B T \ln 2}, \qquad k_B T \ln 2\big|_{T=300\,\mathrm{K}} \approx 2.9{\times}10^{-21}\ \mathrm{J}. \tag{59}$$

Modern ASIC circuits operating at ~ 15–$20\,\mathrm{J/TH}$ yield $E_{\mathrm{hash}} \approx 1.5$–$2 \times 10^{-11}\,\mathrm{J}$, or $\Lambda_{\mathrm{sim}} \sim 10^{10}$. *Weak Prediction:* $\Lambda_{\mathrm{sim}}(t)$ decreases monotonically across successive technological hardware generations. *Limitation:* Observed decreases are consistent with (P2), but are fully explained by standard semiconductor scaling. This test can only *constrain* (P2), never prove it. A persistent plateau far above unity imposed by CMOS thermodynamic limits would render the identification $k_{\mathrm{Nak}} \equiv h$ purely formal in $\mathcal{U}_0$.

7.3. Non-Testable Statements

For epistemic transparency, we designate as *non-falsifiable within the scope of this work*:

1. Postulate (P1) (Bostrom's ancestor simulation hypothesis), which produces no distinct local observables within $\mathcal{U}_0$;
2. The ontological identification of the Timechain with the causal ledger of a parent substrate (P3);
3. Geometric analogies (celestial holography, BMS symmetries, Weyl curvature, halving inflaton), which lack direct observational coupling to astrophysical cosmological datasets (CMB, SNe Ia, BAO);
4. The Societal Trust metric C_s, absent an independent operational measurement procedure outside the model.

Statement	Dependence	Status
T1: Exponential block times	None	Testable
T2: Fixed-point bias	None	Testable
T3: Stale block rate	None	Testable
T4: Fork decay	None	Established
Monotonic $\Lambda_{\mathrm{sim}}(t)$ decay	(P2)	Weak constraint
$k_{\mathrm{Nak}} \equiv h$	(P1)–(P3)	Non-testable
Geometric correspondences	(P1)–(P3)	Heuristic

In summary, the empirical content of this work is restricted to tests T1 through T3 on the real Timechain. The simulation-based interpretation offers a self-consistent conceptual synthesis, but presents no empirical evidence that compels selecting it over standard classical distributed computing theory.

8. CONCLUSION: THE UNIFIED ACTION OF THE TIMECHAIN

Subject to postulates (P1)–(P3), the correspondences developed in this work can be synthesized into an effective unified action functional $\mathcal{S}_{\mathrm{unified}}$ governing the quantum, holographic, thermodynamic, and cosmological dynamics of the Nakamoto consensus:

$$\mathcal{S}_{\text{unified}} = \underbrace{\int_{\mathcal{M}} d^4x\sqrt{-g}\left(\frac{R-2\Lambda_{\text{Nak}}}{16\pi G_{\text{Nak}}} - \frac{1}{2}\partial_\mu\phi\partial^\mu\phi - V_{\text{Halving}}(\phi) + \mathcal{L}^{\text{UTXO}}_{\text{Spinfoam}}\right)}_{\text{Gravitational Bulk, FLRW Cosmology, and UTXO Spin Network}}$$
$$+ \underbrace{\frac{1}{8\pi G_{\text{Nak}}}\oint_{\mathcal{I}^+} du_{\text{ret}}d^2z\sqrt{\gamma}\left(m_B\partial_{u_{\text{ret}}}f + \frac{1}{4}C_{zz}N^{zz} + \mathcal{L}^{\text{Mempool}}_{\text{soft}}\right)}_{\text{BMS Celestial Boundary, Soft Gravitons, and Gravitational Memory}} - \underbrace{k_{\text{Nak}}\int_{\mathcal{C}_{a+ib}} dz\ \text{Tr}\left(\mathcal{L}_{\text{GKSL}}[\rho]\ln\rho\right)}_{\text{Complex KMS Thermal Time } (a+ib),\ \text{GKSL Decoherence, and}} \tag{60}$$

This unified action illustrates that Satoshi Nakamoto's architecture extends beyond applied computer science: by anchoring digital information in irreversible thermodynamic dissipation ($k_{\text{Nak}} \equiv h$), the Timechain provides a discrete toy model of a quantum, relativistic universe. In this framework, time emerges from thermal dissipation (Connes-Rovelli), value conservation stems from celestial sphere asymptotic symmetries (Pasterski-Strominger-Weinberg), sequential block creation expands the fabric of spacetime (Sorkin-Padmanabhan-Susskind), and economic curvature (C_s, u) reflects the horizon equation of state (Jacobson).

From this perspective, Nakamoto consensus physically realizes what John F. Nash Jr. conceptualized game-theoretically as *Ideal Money* [21]: an incorruptible monetary standard whose asymptotic stability ($u \to u^*_{\text{Nash}}$) is guaranteed not by human institutions, but by the fundamental conservation laws and entropy of a self-regulating informational spacetime.

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